

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Phosphorus trichloride

**CAS No** 7719-12-2

**Synonyms** Phosphoric chloride.; Phosphorus perchloride

**Product Code** **C16948**

**Address** ThermoFisher Scientific Australia Pty Ltd  
5 Caribbean Drive, Scoresby  
VICTORIA 3179, Australia

**Emergency Tel.** **CHEMTREC®**  
**03 9757 4559 or +613 9757 4559**

**Telephone / Fax Numbers** Tel: 1300 735 292  
Fax: 1800 067 639

**E-mail address** ANZinfo@thermofisher.com

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product contains one or more substance(s) on the Illicit Drug Precursors/Reagents list. Verify requirements related to using, handling and storing these substances. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product contains one or more substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

**Physical hazards**  
No hazards identified

### **Health hazards**

|                                                      |              |
|------------------------------------------------------|--------------|
| Acute Oral Toxicity                                  | Category 2   |
| Acute Inhalation Toxicity - Vapors                   | Category 2   |
| Skin Corrosion/Irritation                            | Category 1 A |
| Serious Eye Damage/Eye Irritation                    | Category 1   |
| Specific target organ toxicity - (repeated exposure) | Category 2   |

**Environmental hazards**  
No hazards identified

### **Label Elements**



Skull and Crossbones



Health Hazard



Corrosion

**Signal Word****Danger****Hazard Statements**

H314 - Causes severe skin burns and eye damage

H373 - May cause damage to organs through prolonged or repeated exposure

H300 + H330 - Fatal if swallowed or if inhaled

AUH014 - Reacts violently with water

AUH029 - Contact with water liberates toxic gas

**Precautionary Statements**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

P271 - Use only outdoors or in a well-ventilated area

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P284 - Wear respiratory protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P310 - Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

P331 - Do NOT induce vomiting

P363 - Wash contaminated clothing before reuse

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

P405 - Store locked up

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

No information available

**Section 3 - Composition and Information on Ingredients**

| Component              | CAS No    | Weight % |
|------------------------|-----------|----------|
| Phosphorus trichloride | 7719-12-2 | >95      |

**Section 4 - First Aid Measures****Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

|                                            |                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                        |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                         |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                     |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: After inhalation exposure, observe for 24 to 72 hours as pulmonary edema may be delayed |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                                                   |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

### Extinguishing media which must not be used for safety reasons

Water.

### Hazardous Decomposition Products

Oxides of phosphorus, Hydrogen chloride gas.

### Specific Hazards Arising from the Chemical

Contact with water liberates toxic gas. Water reactive. Produce flammable gases on contact with water.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas.

### Environmental Precautions

Prevent further leakage or spillage if safe to do so.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Wear self-contained breathing apparatus and protective suit. Do not expose spill to water.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling**

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not allow contact with water. Wash hands before breaks and immediately after handling the product.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Protect from moisture. Corrosives area. Keep under nitrogen. Keep containers tightly closed in a dry, cool and well-ventilated place.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

## Section 8 - Exposure Controls and Personal Protection

**Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component              | Australia                                                                                  | New Zealand WEL                                                                            | ACGIH TLV                     | The United Kingdom                                                                                                 | Germany                                                                                                                                                                                                                                                           |
|------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phosphorus trichloride | STEL: 0.5 ppm<br>STEL: 2.8 mg/m <sup>3</sup><br>TWA: 0.2 ppm<br>TWA: 1.1 mg/m <sup>3</sup> | TWA: 0.2 ppm<br>TWA: 1.1 mg/m <sup>3</sup><br>STEL: 0.5 ppm<br>STEL: 2.8 mg/m <sup>3</sup> | TWA: 0.2 ppm<br>STEL: 0.5 ppm | STEL: 0.5 ppm 15 min<br>STEL: 2.9 mg/m <sup>3</sup> 15 min<br>TWA: 0.2 ppm 8 hr<br>TWA: 1.1 mg/m <sup>3</sup> 8 hr | TWA: 0.1 ppm (8 Stunden). AGW - exposure factor 1<br>TWA: 0.57 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 0.1 ppm (8 Stunden). MAK<br>TWA: 0.57 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.1 ppm<br>Höhepunkt: 0.57 mg/m <sup>3</sup> |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |
| Nitrile rubber |                                   |                 |                 |                       |
| Neoprene       |                                   |                 |                 |                       |
| PVC            |                                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Wear appropriate protective gloves and clothing to prevent skin exposure                                                                                                                                                                                                                                                    |
| <b>Respiratory Protection</b>          | Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices |
| <b>Recommended Filter type:</b>        | Particulates filter conforming to EN 143 Inorganic gases and vapours filter Type B Grey conforming to EN14387 (or AUS/NZ equivalent)                                                                                                                                                                                        |
| <b>Recommended half mask:-</b>         | Particle filtering: EN149:2001 (or AUS/NZ equivalent)<br>When RPE is used a face piece Fit Test should be conducted                                                                                                                                                                                                         |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                                                                      |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                                                                   |

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                                |                               |                                          |
|------------------------------------------------|-------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Clear                         |                                          |
| <b>Physical State</b>                          | Liquid                        |                                          |
| <b>Odor</b>                                    | pungent                       |                                          |
| <b>Odor Threshold</b>                          | No data available             |                                          |
| <b>pH</b>                                      | 1                             | 5 g/l aq.sol                             |
| <b>Melting Point/Range</b>                     | -112 °C / -169.6 °F           |                                          |
| <b>Softening Point</b>                         | No data available             |                                          |
| <b>Boiling Point/Range</b>                     | 74 - 78 °C / 165.2 - 172.4 °F |                                          |
| <b>Flash Point</b>                             | No information available      | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available             |                                          |
| <b>Flammability (solid,gas)</b>                | Not applicable                | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available             |                                          |
| <b>Vapor Pressure</b>                          | 115 mmHg @ 25 °C              |                                          |
| <b>Vapor Density</b>                           | 4.74                          | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | 1.570                         |                                          |
| <b>Bulk Density</b>                            | Not applicable                | Liquid                                   |
| <b>Water Solubility</b>                        | reacts                        |                                          |
| <b>Solubility in other solvents</b>            | No information available      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                               |                                          |
| <b>Autoignition Temperature</b>                | No data available             |                                          |
| <b>Decomposition Temperature</b>               | No data available             |                                          |
| <b>Viscosity</b>                               | No data available             |                                          |
| <b>Explosive Properties</b>                    | No information available      |                                          |
| <b>Oxidizing Properties</b>                    | No information available      |                                          |
| <b>Other information</b>                       |                               |                                          |
| <b>Molecular Formula</b>                       | Cl <sub>3</sub> P             |                                          |
| <b>Molecular Weight</b>                        | 137.33                        |                                          |

## Section 10 - Stability and Reactivity

|                                         |                                                                          |
|-----------------------------------------|--------------------------------------------------------------------------|
| <b>Reactivity</b>                       | Yes                                                                      |
| <b>Stability</b>                        | Water reactive. Contact with water liberates toxic gas. Light sensitive. |
| <b>Conditions to Avoid</b>              | Incompatible products, Exposure to moist air or water.                   |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Finely powdered metals, Metals.                 |
| <b>Hazardous Decomposition Products</b> | Oxides of phosphorus. Hydrogen chloride gas.                             |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                 |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                            |                                                                  |
|----------------------------|------------------------------------------------------------------|
| <b>(a) acute toxicity;</b> |                                                                  |
| <b>Oral</b>                | Category 2                                                       |
| <b>Dermal</b>              | Based on available data, the classification criteria are not met |
| <b>Inhalation</b>          | Category 2                                                       |

| Component              | LD50 Oral      | LD50 Dermal                     | LC50 Inhalation  |
|------------------------|----------------|---------------------------------|------------------|
| Phosphorus trichloride | 18 mg/kg (Rat) | LD50 250 - 500 mg/kg ( Rabbit ) | 104 ppm/4h (Rat) |

|                                       |              |
|---------------------------------------|--------------|
| <b>(b) skin corrosion/irritation;</b> | Category 1 A |
|---------------------------------------|--------------|

|                                           |                                                                  |
|-------------------------------------------|------------------------------------------------------------------|
| <b>(c) serious eye damage/irritation;</b> | Based on available data, the classification criteria are not met |
|-------------------------------------------|------------------------------------------------------------------|

|                                               |                                                                  |
|-----------------------------------------------|------------------------------------------------------------------|
| <b>(d) respiratory or skin sensitization;</b> |                                                                  |
| <b>Respiratory</b>                            | Based on available data, the classification criteria are not met |
| <b>Skin</b>                                   | Based on available data, the classification criteria are not met |

|                                    |                                                                  |
|------------------------------------|------------------------------------------------------------------|
| <b>(e) germ cell mutagenicity;</b> | Based on available data, the classification criteria are not met |
|------------------------------------|------------------------------------------------------------------|

Not mutagenic in AMES Test

|                             |                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>(f) carcinogenicity;</b> | Based on available data, the classification criteria are not met<br>There are no known carcinogenic chemicals in this product |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|

|                                   |                                                                  |
|-----------------------------------|------------------------------------------------------------------|
| <b>(g) reproductive toxicity;</b> | Based on available data, the classification criteria are not met |
|-----------------------------------|------------------------------------------------------------------|

|                                  |                                                                  |
|----------------------------------|------------------------------------------------------------------|
| <b>(h) STOT-single exposure;</b> | Based on available data, the classification criteria are not met |
|----------------------------------|------------------------------------------------------------------|

|                                    |                     |
|------------------------------------|---------------------|
| <b>(i) STOT-repeated exposure;</b> | Category 2          |
| <b>Route of exposure</b>           | Inhalation          |
| <b>Target Organs</b>               | Respiratory system. |

|                               |                                                                  |
|-------------------------------|------------------------------------------------------------------|
| <b>(j) aspiration hazard;</b> | Based on available data, the classification criteria are not met |
|-------------------------------|------------------------------------------------------------------|

|                                                   |                                                                                                                                                                                                                                                                   |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Symptoms / effects, both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: After |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

inhalation exposure, observe for 24 to 72 hours as pulmonary edema may be delayed

## Section 12 - Ecological Information

### Ecotoxicity effects

Do not empty into drains. .

| Component              | Freshwater Fish                             | Water Flea | Freshwater Algae | Microtox |
|------------------------|---------------------------------------------|------------|------------------|----------|
| Phosphorus trichloride | LC50: > 1000 mg/L, 96h static (Danio rerio) |            |                  |          |

### Persistence and Degradability

#### Persistence

Persistence is unlikely, based on information available.

#### Degradability

Not relevant for inorganic substances.

#### Bioaccumulative Potential

Bioaccumulation is unlikely

### Mobility

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility Disperses rapidly in air

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

### Persistent Organic Pollutant

This product does not contain any known or suspected substance

### Ozone Depletion Potential

This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

### Waste from Residues/Unused Products

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

### Contaminated Packaging

Dispose of this container to hazardous or special waste collection point.

### Other Information

Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms. Solutions with low pH-value must be neutralized before discharge.

## Section 14 - Transport Information

### IMDG/IMO

UN-No UN1809  
Proper Shipping Name PHOSPHORUS TRICHLORIDE  
Hazard Class 6.1  
Subsidiary Hazard Class 8  
Packing Group I

### ADG

UN-No UN1809  
Proper Shipping Name PHOSPHORUS TRICHLORIDE  
Hazard Class 6.1  
Subsidiary Hazard Class 8  
Packing Group I

| Component                                  | Hazchem Code |
|--------------------------------------------|--------------|
| Phosphorus trichloride<br>7719-12-2 (>95 ) | 4WE          |

### IATA

FORBIDDEN FOR IATA TRANSPORT

UN-No UN1809  
Proper Shipping Name PHOSPHORUS TRICHLORIDE, FORBIDDEN FOR IATA TRANSPORT  
Hazard Class 6.1  
Subsidiary Hazard Class 8  
Packing Group I

Environmental hazards No hazards identified  
Special Precautions No special precautions required  
Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

No poison schedule number allocated.

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                          | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|------------------------------------|-------------------------------------------------------------|------------------------|
| Phosphorus trichloride - 7719-12-2 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product contains one or more substance(s) on the Illicit Drug Precursors/Reagents list. Verify requirements related to using, handling and storing these substances.

#### **Chemicals of Security Concern**

This product contains one or more substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

| Component                          | Australian - Illicit Drug Precursors/Reagents Substance List | Chemicals of Security Concern |
|------------------------------------|--------------------------------------------------------------|-------------------------------|
| Phosphorus trichloride - 7719-12-2 | Category 3                                                   | Listed in Appendix A          |

#### **Legend**

Category 3 - Chemicals and apparatus that may be used in the illicit production of drugs. Purchases from this list should alert companies or organizations to seek further indicators of any suspicious orders or enquiries. No official reporting is required for items on this list unless considered warranted

**Chemicals of Security Concern** - for further information see <http://www.chemicalsecurity.gov.au/securityconcerns>

**National pollutant inventory** Not applicable

#### **Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.



**International Inventories**

| Component              | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|------------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Phosphorus trichloride | X    | X     | 231-749-3 | -      | X    | X   | -    | X     | X    | X    | X     | KE-28723 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**  
Not applicable.

| Component              | CAS No    | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|------------------------|-----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Phosphorus trichloride | 7719-12-2 | Listed   | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

**Authorisation/Restrictions according to EU REACH**

| Component              | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Phosphorus trichloride | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Section 16 - Other Information****Legend**

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZIoC** - New Zealand Inventory of Chemicals  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

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**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

#### Revision Date

20-Nov-2022

#### Revision Summary

Initial Release.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

#### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**